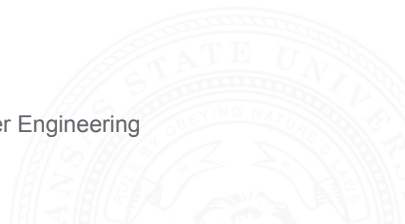


Three-Phase Transformers II

Three-phase network models, autotransformers, and taps

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Lecture outline

- 1 Build a balanced three-phase model from the single-phase equivalent.
- 2 A complete bank-rating and impedance example.
- 3 Autotransformer construction and operating concept.
- 4 Physical taps, off-nominal ratio, and model conventions.

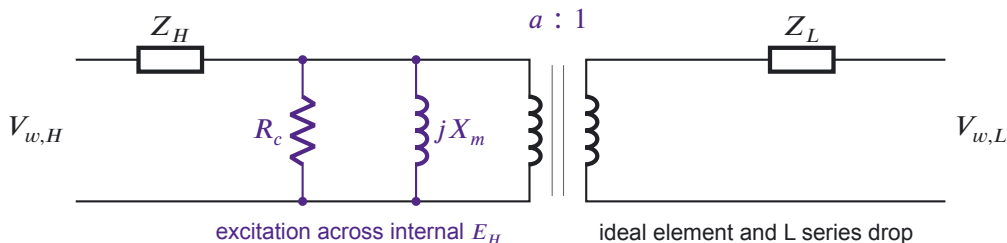
Use the winding connections and ratings to build the balanced network model.

PART 1

From single-phase to three-phase model

Quick recap: single-phase transformer

Single-phase unit: V_w is coil terminal voltage; $a = N_H/N_L$.



Three impedance parts

H series: $Z_H = R_H + jX_{\sigma,H}$.

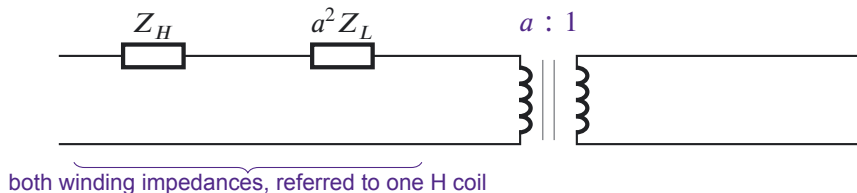
L series: $Z_L = R_L + jX_{\sigma,L}$.

Excitation: $R_c \parallel jX_m$ (core loss and magnetization).

Two winding series impedances, an excitation branch, and an ideal turns-ratio element.

Quick recap: reduced series model

Loaded operation: neglect I_0 when it is small relative to load current.



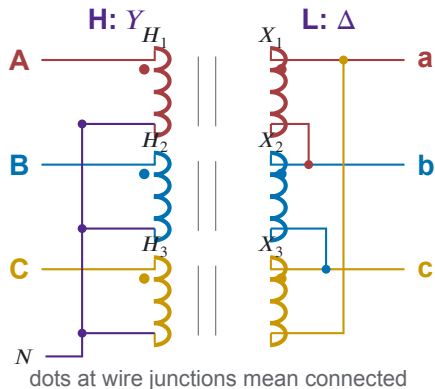
$$Z_H + a^2 Z_L = (R_H + a^2 R_L) + j(X_{\sigma,H} + a^2 X_{\sigma,L})$$

This is the reduced equivalent of one single-phase unit.

Retain $R_c \parallel jX_m$ when excitation or core loss is required.

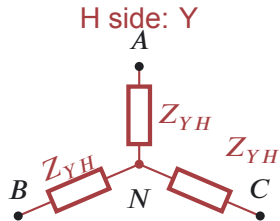
First choose the approximation; then refer and combine the series impedances.

Three-phase Y- Δ connection



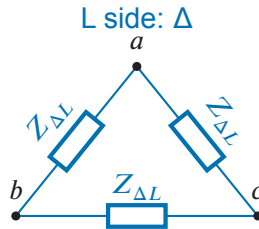
Series impedance on H or L

Balanced model: retain resistance and leakage; neglect excitation.



$$Z_{YH} = Z_H + a^2 Z_L$$

Combined series impedances (Ω): H or L referral



$$Z_{\Delta L} = Z_L + Z_H/a^2$$

Next: replace the L Δ impedance network with an equivalent Y.

L-side $\Delta \rightarrow Y$ equivalent



$$Z_{YL} = \frac{Z_{\Delta L}}{3}$$

n : equivalent L neutral reference; no grounding implied.

The impedance representation changes; the transformer stays Y- Δ .

Phase voltages and displacement

Positive-sequence phases

$$V_{AN,H} = \frac{V_{AB,H}}{\sqrt{3}e^{j30^\circ}}$$

$$V_{an,L} = \frac{V_{ab,L}}{\sqrt{3}e^{j30^\circ}}$$

N : physical H winding neutral.

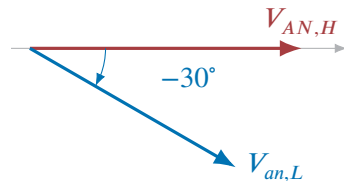
n : equivalent L neutral reference.

$$\delta_{LH} = \angle V_{an,L} - \angle V_{AN,H} = -30^\circ$$

For this ideal joining, $V_{AN,H}$ leads $V_{an,L}$ by 30° .

Ideal Y- Δ joining

AN-ab pair: $V_{ab,L} = V_{AN,H}/a$.



phasor lengths scaled for display

Voltage normalization

Line-to-line voltage bases in the rated terminal ratio:

$$\frac{V_{B,H}}{V_{B,L}} = \frac{V_{LL,H,\text{rated}}}{V_{LL,L,\text{rated}}} = \sqrt{3}a$$

H phase voltage (pu)

L phase voltage (pu)

$$v_H = \frac{V_{AN,H}}{V_{B,H}/\sqrt{3}}$$

$$v_L = \frac{V_{an,L}}{V_{B,L}/\sqrt{3}}$$

$$v_H = e^{j\theta} v_L \text{ (ideal),} \quad \theta = -\delta_{LH} = +30^\circ$$

Compatible bases absorb the magnitude ratio and retain the connection angle.

Phase impedance normalization

Resistance and leakage: Z_{YH} is the H-side phase impedance (Ω).

Compatible system bases

$V_{B,H}, V_{B,L}$: line-to-line kV, in the rated ratio.

S_B : total three-phase MVA; each phase uses $S_B/3$.

$$Z_{B,H} = \frac{(V_{B,H}/\sqrt{3})^2}{S_B/3} = \frac{V_{B,H}^2}{S_B}$$

$$z_{eq} = \frac{Z_{YH}}{Z_{B,H}}$$

Line-to-line kV and total three-phase MVA give the phase impedance base.

Current normalization

Use terminal line-current bases on both voltage sides.

H current base

$$I_{B,H} = \frac{S_B}{\sqrt{3}V_{B,H}}$$

$$i_H = \frac{I_A}{I_{B,H}}$$

I_A enters the H terminal.

S_B : total three-phase MVA; $V_{B,H}, V_{B,L}$: line-to-line kV.

The equivalent L phase branch carries line current I_a .

L current base

$$I_{B,L} = \frac{S_B}{\sqrt{3}V_{B,L}}$$

$$i_L = \frac{I_a}{I_{B,L}}$$

I_a leaves the L terminal.

Normalize terminal currents first; use power balance next to obtain their phase relation.

Power normalization

Physical total power

$$S_{3\phi} = 3V_{\phi}I_{\text{line}}^*$$

V_{ϕ} : terminal phase voltage.

Per-unit bank power

$$s = \frac{S_{3\phi}}{S_B} = \frac{3V_{\phi}I_{\text{line}}^*}{3V_{B,\phi}I_B} = vi^*$$

Phase bases: $V_{B,\phi}I_B = S_B/3$.

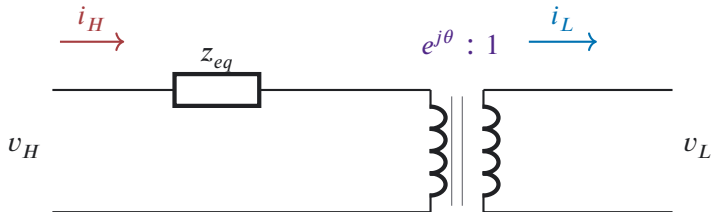
Ideal element: complex power conserved

Use $v_H = e^{j\theta}v_L$, $\theta = +30^\circ$; series drop and excitation omitted.

$$v_H i_H^* = v_L i_L^* \implies i_L = e^{-j\theta} i_H$$

The factor 3 cancels in pu power; ideal power balance fixes the current phase relation.

Assemble the balanced Y-Δ branch



balanced positive sequence; excitation neglected; resistance and leakage retained

$$v_H = \underbrace{z_{eq} i_H}_{\text{series drop}} + \underbrace{e^{j\theta} v_L}_{\text{ideal connection}}, \quad i_L = e^{-j\theta} i_H$$
$$z_{eq} = r_{eq} + jx_{eq}; \theta = +30^\circ; s_H - s_L = z_{eq} |i_H|^2 \text{ (all in pu).}$$

The final branch retains both winding series effects and the connection displacement.

Four connections: network summary

Balanced positive sequence, excitation neglected. $a = N_H/N_L$.

H-L	Ideal line ratio $ V_{LL,H} / V_{LL,L} $	H-side phase impedance $Z_{YH} (\Omega)$	H-from-L angle $\theta = -\delta_{LH}$
Y-Y	a	$Z_H + a^2 Z_L$	0°
Y- Δ	$\sqrt{3}a$	$Z_H + a^2 Z_L$	$+30^\circ$
Δ -Y	$a/\sqrt{3}$	$Z_{\Delta H}/3$	$+30^\circ$
Δ - Δ	a	$Z_{\Delta H}/3$	0°

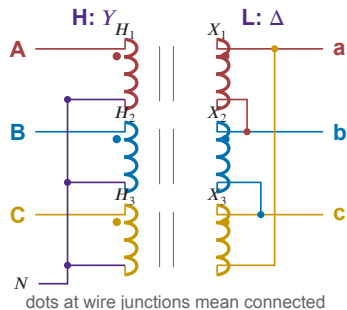
$Z_{\Delta H} = Z_H + a^2 Z_L$. Mixed-connection angles use the H-leading-L convention.

All four connections use the same balanced branch structure.

PART 2

Bank example and checks

Bank example: data



Equipment and system bases

60 MVA, 138/13.8 kV, Y-Δ.

$$z_{60} = 0.006 + j0.090 \text{ pu}$$

$$S_B = 100 \text{ MVA}$$

$$V_{B,H} = 138 \text{ kV}$$

$$V_{B,L} = 13.8 \text{ kV}$$

Paired coils: (AN, ab), (BN, bc), (CN, ca).

Positive sequence; H neutral N available.

Grounding method is unspecified.

The kV ratings are line-to-line voltages; 60 MVA is the total bank rating.

Bank example: solution route

- | | |
|-----------------------|---|
| 1. Given | Rated terminal kV, bank MVA, coil joining, and series-impedance data. |
| 2. Model | Balanced positive-sequence branch; excitation neglected, resistance and leakage retained. |
| 3. Calculation | Recover coil ratio; derive L phasor; convert the combined series impedance. |
| 4. Check | Check actual ohms and rated currents independently. |

Use physical quantities to check the normalized model.

Bank example: voltage ratios

Coil voltages

$$E_H = 138/\sqrt{3} = 79.6743 \text{ kV}$$

$$E_L = 13.8 \text{ kV}$$

$$a = 10/\sqrt{3} = 5.77350$$

Terminal voltages

$$\frac{V_{LL,H,\text{rated}}}{V_{LL,L,\text{rated}}} = \frac{138}{13.8} = 10$$

$$\frac{V_{LL,H,\text{rated}}}{V_{LL,L,\text{rated}}} = \sqrt{3}a$$

Both ratios describe the same bank.

The Y-Δ coil ratio is the terminal line ratio divided by $\sqrt{3}$.

Bank example: no-load phasor

Reference: $V_{AB,H} = 138\angle 0^\circ$ kV.

$$V_{AN,H} = \frac{V_{AB,H}}{\sqrt{3}e^{j30^\circ}} = 79.6743\angle(-30^\circ) \text{ kV}$$

$$\begin{aligned} V_{ab,L} &= V_{AN,H}/a \\ &= 11.95115 - j6.90000 = 13.800\angle(-30^\circ) \text{ kV} \end{aligned}$$

$$V_{an,L} = \frac{V_{ab,L}}{\sqrt{3}e^{j30^\circ}} = 7.96743\angle(-60^\circ) \text{ kV}$$

$V_{an,L}$ uses the equivalent L neutral reference.

Transfer the AN–ab coil voltage, then form the equivalent L phase voltage.

Bank example: change of base

$$z_{new} = z_{old} \frac{S_{B,new}}{S_{B,old}} \left(\frac{V_{B,old}}{V_{B,new}} \right)^2$$

The voltage bases on each physical side are unchanged.

$$z_{100} = (0.006 + j0.090) \frac{100}{60} = 0.010 + j0.150 \text{ pu}$$

The per-unit number changes because the impedance base changes; the physical impedance stays fixed.

Bank example: recover ohms

Side	$Z_{B,60} (\Omega)$	$Z_Y (\Omega)$
H (Y)	$138^2/60 = 317.4$	$1.9044 + j28.566$
L (equiv. Y)	$13.8^2/60 = 3.174$	$0.019044 + j0.28566$

$$\frac{Z_{YH}}{Z_{YL}} = \left(\frac{V_{LL,H,\text{rated}}}{V_{LL,L,\text{rated}}} \right)^2 = 100$$

On the 100-MVA system base, $Z_{B,H} = 190.44 \Omega$ and $Z_{B,L} = 1.9044 \Omega$.

L Δ branches: $Z_{\Delta L} = 3Z_{YL} = 0.057132 + j0.85698 \Omega$.

Either MVA base reproduces the same physical terminal equivalent.

Bank example: rated currents

Equipment currents

$$I_{H,r} = \frac{60}{\sqrt{3}(138)} = 0.251022 \text{ kA}$$

$$I_{L,r} = \frac{60}{\sqrt{3}(13.8)} = 2.51022 \text{ kA}$$

Coil-current magnitudes:

$$I_{w,H} = I_{H,r} \text{ (Y).}$$

$$I_{w,L} = I_{L,r}/\sqrt{3} = 1.44928 \text{ kA } (\Delta).$$

System-base check

$$I_{H,B} = 0.41837 \text{ kA}$$

$$I_{L,B} = 4.18370 \text{ kA}$$

Rated current is 0.60 pu on the 100-MVA base.

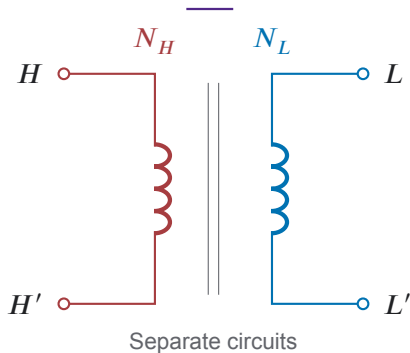
Rated equipment current is 1 pu on its own MVA base and 0.60 pu on the system base.

PART 3

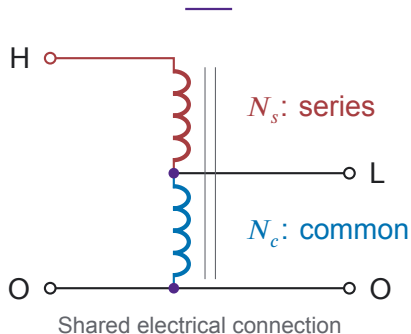
Autotransformer concept

Two-winding and autotransformer

Two-winding transformer

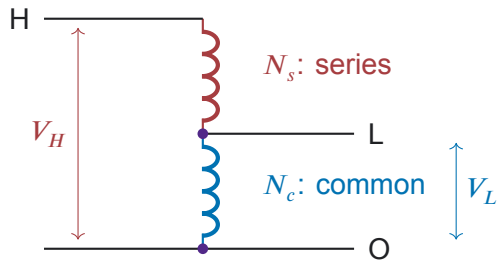


Autotransformer



An autotransformer couples H and L electrically as well as magnetically.

Series and common sections



Series-aiding connection

H is connected across $N_s + N_c$ turns. L is connected across N_c turns.

$$V_H = E_s + E_c, \quad V_L = E_c$$

$$\frac{V_H}{V_L} = \frac{N_s + N_c}{N_c}$$

The common section belongs to both terminal circuits.

Source: Glover et al., sixth edition (2017), Fig. 3.23(b)

Autotransformer power paths

For the shown ideal step-down connection, let I_H enter H and I_L leave L.

$$I_c = I_H - I_L$$

Here I_c is referenced from L toward O through the common section.

$$V_H I_H^* = V_L I_L^*$$

Part of the terminal power is transferred by magnetic coupling; part passes through the direct electrical connection.

A common-winding current is not automatically equal to either terminal current.

Source: Glover et al., sixth edition (2017), Fig. 3.23(b), Sec. 3.7

Autotransformer applications

Why use it?

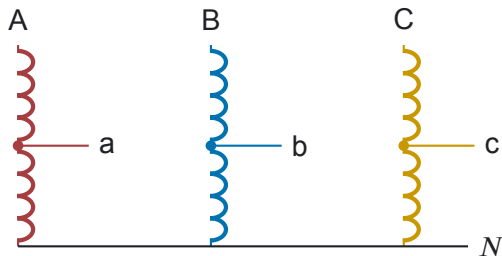
For suitable voltage ratios, less winding material and a smaller magnetic unit can serve the terminal MVA. Common in voltage-level interconnection and voltage adjustment.

What changes?

H and L have no galvanic isolation. Lower series impedance can increase fault current. Grounding and transferred overvoltages need attention.

Its compact construction comes with an electrical connection between the two voltage systems.

Three-phase autotransformer



illustrated common-neutral Y-connected implementation

Each phase has a series section and a common section.

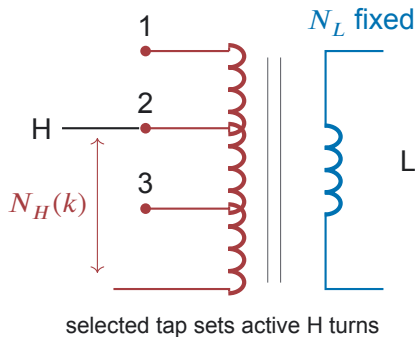
Three-phase construction repeats the autoconnected winding path on all three phases.

Source: Course-redrawn conceptual three-phase extension of textbook Fig. 3.23(b).

PART 4

Tap ratio concept

Tapped winding construction



A selected connection point

Changing position k changes the active turns $N_H(k)$.

$$a(k) = \frac{N_H(k)}{N_L}$$

The Y/ Δ connections remain as specified. A tapped isolated transformer still has separate H and L circuits.

A winding tap changes active turns without necessarily creating an autotransformer.

Source: Course-redrawn conceptual tap circuit. Reinhausen, Tap-changer basics, official description of active-turn selection.

Three different ratios

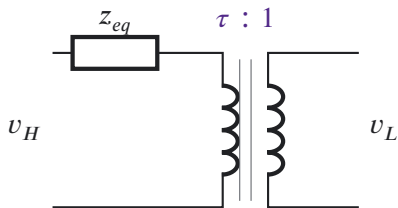
- 1 Physical winding ratio: $a(k) = N_H(k)/N_L$; k labels tap position.
- 2 Actual line-voltage ratio: $|V_{LL,H}(k)|/|V_{LL,L}(k)|$, from $a(k)$ and Y/ Δ joining.
- 3 Voltage-base ratio: $b = V_{B,H}/V_{B,L}$, fixed by the system model.

$$\tau(k) = \frac{|V_{LL,H}(k)|/|V_{LL,L}(k)|}{V_{B,H}/V_{B,L}}$$

With bases in the rated line-voltage ratio, the nominal tap has $\tau = 1$.

The off-nominal per-unit tap is actual terminal ratio divided by the chosen base ratio.

Off-nominal model



fixed bases; exciting branch
and phase shift omitted

Textbook reduction

a_t : equivalent terminal ratio; b : selected base ratio.

$$c = \frac{a_t}{b}$$

With phase shift omitted, our magnitude notation is $c = \tau$.

$$\tau = \frac{|V_{LL,H}|}{|V_{LL,L}|} \frac{1}{b}$$

Compatible nominal bases do not absorb a tap-dependent change in the actual ratio.

Source: Glover et al., sixth edition (2017), Fig. 3.25(a)–(c), $c = a_t/b$

Tap magnitude and voltage

Define the ideal relation in this direction:

$$v_H = \tau e^{j\theta} v_L$$

For fixed v_H and negligible leakage drop,

$$|v_L| = \frac{|v_H|}{\tau}$$

Increasing active H turns increases the H/L line-voltage ratio and τ , so the L voltage falls in this stated case.

“Raise” or “lower” requires the tapped side, controlled voltage, and defining ratio.

Tap-changing equipment

De-energized tap changer

Tap position is changed with the transformer de-energized. Sets a planned ratio for the expected operating conditions.

On-load tap changer

Changes ratio while energized and carrying load. A designed transition path prevents interruption of load current and an unimpeded short between taps.

A steady tap diagram does not show the switching mechanism needed for on-load operation.

Magnitude and phase in one model

$$t_{HL} = \tau e^{j\theta}, \quad v_H = t_{HL} v_L \quad (\text{ideal element})$$

$$\theta = -\delta_{LH}, \quad t_{LH} = t_{HL}^{-1}$$

Ordinary ratio taps adjust τ . The connection phase displacement remains fixed for the same terminal joining. A phase-shifting transformer is a distinct equipment function.

Record the defining equation and direction together with every complex ratio.

Summary

- 1 Choose the model level; retain winding resistance and leakage, and state the excitation assumption.
- 2 Use the connection to obtain phase impedance; retain voltage and current displacement.
- 3 An autotransformer has shared series/common sections and no H/L galvanic isolation.
- 4 An off-nominal tap is the actual ratio divided by the chosen base ratio.

A useful transformer model preserves the physical connection and names every ratio.